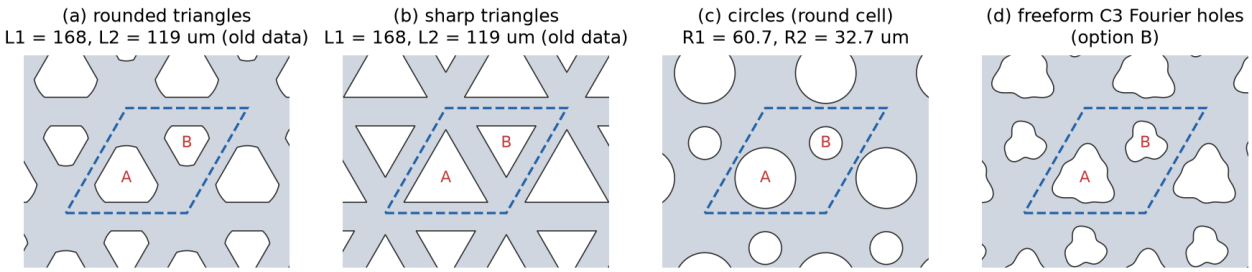
AI-Assisted Inverse Design of Terahertz Valley Photonic Crystals

Study guide: freeform hole shapes (option B), the data, the physics, the code, the results

M.Tech project P29 — Binit Shakya, IIT Guwahati
Supervisors: Dr. Debabrata Sikdar, Dr. Teena Sharma

Prepared 28 October 2026, for the first-week-of-November evaluation



The four kinds of unit cell this guide talks about. Grey = silicon, white = air holes, dashed rhombus = one unit cell with its two holes A and B.

Contents

1. The project in one page 4

2. Physics you need 6

2.1 Photonic crystals and band diagrams . . . 6

2.2 The lattice, the two holes and the k-path . . . 6

2.3 Dirac cones and the valley-Hall effect . . . 6

2.4 Two useful invariances . . . 7

3. The data you have: the COMSOL datasets 8

3.1 What COMSOL computed . . . 8

3.2 Why 29.5 million rows are only 128 designs . . . 8

3.3 The exact hole shapes (worked out for option B) . . . 8

3.4 Where the band edges sit . . . 8

3.5 The round-cell dataset and the k = 1, 2 re-run . . . 9

4. Stage 1 recap: the L1/L2 benchmark 10

5. Option B: describing any C3 hole with a few numbers 11

5.1 The symmetry selection rule . . . 11

5.2 One descriptor for every dataset . . . 11

5.3 The design space for inverse design . . . 12

5.4 Fabrication and validity rules . . . 12

5.5 Different designs that are physically the same . . . 12

6. A fast simulator: plane-wave expansion (PWE) 13

6.1 Why . . . 13

6.2 How it works . . . 13

6.3 Checking it against the COMSOL datasets . . . 13

6.4 Why COMSOL needs only k = 1 (M) and k = 2 (K) . . . 14

7. The option-B dataset 17

8. Forward surrogate: shape → band edges 18

8.1 The model . . . 18

8.2 How many simulations are needed? . . . 18

8.3 Accuracy of the final ensemble . . . 18

9. Using the COMSOL datasets: multi-fidelity correction 20

9.1 Freeform shapes in COMSOL . . . 20

10. Inverse design: from target band edges to shapes 22

10.1 The problem . . . 22

10.2 Method 1 — tandem network . . . 22

10.3 Method 2 — conditional variational autoencoder (CVAE) . . . 22

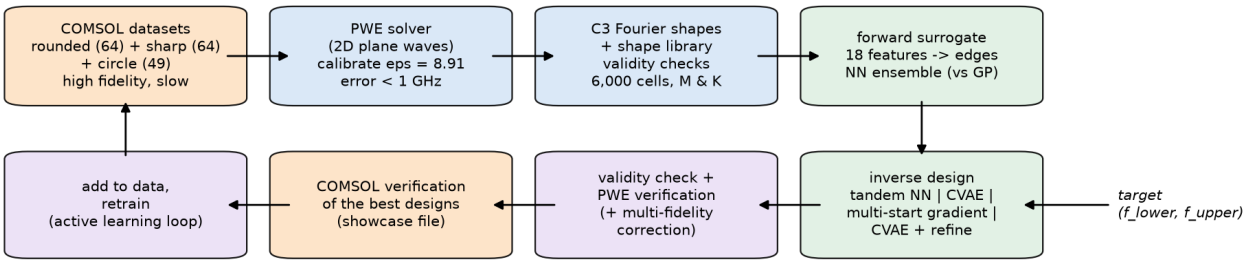
10.4 Method 3 — multi-start gradient search . . . 22
10.5 Method 4 — CVAE + refine . . . 22
10.6 How the methods were judged . . . 22
10.7 What the comparison shows . . . 23
10.8 The design gallery: many shapes, one target . . . 25
11. How to run everything, and the COMSOL side **26**
11.1 Commands . . . 26
11.2 Building a Fourier-hole model in COMSOL . . . 26
11.3 Verifying the gallery designs . . . 26
11.4 Adding a new COMSOL sweep (how the round-cell data went in) . . . 26
11.5 Code map (src/shapes) . . . 27
12. Limitations — say these before an examiner does **28**
13. What to do next **28**
14. Glossary **29**
15. Test yourself **29**
16. References **31**

1. The project in one page

Goal. 6G links will use carrier frequencies around 300 GHz (the IEEE 802.15.3d band is 252.72–321.84 GHz). On a chip, terahertz waves can be guided with almost no loss and around sharp bends by *topological* waveguides made from **valley photonic crystals (VPCs)**: a silicon slab with a periodic pattern of air holes. The holes decide the **photonic band gap** — the frequency range in which no wave can travel through the crystal — and the topological edge channel lives inside that gap. So the design question is:

The inverse-design question

Given a target band gap (lower edge f_{lower}, upper edge f_{upper}, in GHz), **what hole shapes produce it?** The usual way is trial and error in COMSOL. The project replaces that with machine learning that goes directly from target to shape.



The option-B pipeline. Orange = COMSOL (slow, accurate), blue = fast solver and data, green = machine learning, purple = checks and the loop back.

Where the project was. Until now the hole shape was fixed (a triangle with rounded corners) and only its two side lengths L1, L2 were designed, from 64 COMSOL simulations. With 2 inputs and 2 outputs that problem turned out to be one-to-one, so a plain Gaussian-process fit beat every neural network (chapter 4).

What option B adds. The hole may now have *any* shape, as long as it keeps the three-fold symmetry the physics needs. Every shape — circle, rounded or sharp triangle, rotated triangle, hexagon-like, trefoil, or a mix with a different shape in each hole — is written as the same list of 18 numbers (a Fourier description of the hole outline). Now many different shapes give the same band gap, which is exactly the situation the tandem network and generative models were invented for.

What was built and found (details in the chapters)

- **The real hole geometry** of the old COMSOL data, worked out from the group's slides and confirmed by simulation: the rounded hole is a triangle of side L *intersected with a disk* of radius 2L/(3√3) centred on the hole (chapter 3).
- **A fast band solver** (plane-wave expansion, ~0.15 s per shape instead of COMSOL's seconds-to-minutes) that reproduces both triangle COMSOL datasets to within about 1 GHz once a single number is fitted: the silicon permittivity, ε = 8.91 (not 11.7). It then predicts the new circle dataset, which it was never fitted on, just as well (MAE 0.92 / 0.39 GHz). That strongly suggests the COMSOL 2D model uses an effective index of n ≈ {eps ** 0.5:.2f} for the 200 µm slab — please confirm in COMSOL (chapter 6).
- **The 2-k-point shortcut still holds** for freeform shapes: in every one of several hundred random shapes the band edges sit exactly at the M or K point, checked against the full k-path and a 2D grid over the Brillouin zone. So COMSOL needs only 2 eigen-solves per shape (chapter 6).
- **A new dataset** of 6,000 fabricable unit cells (4,000 freeform Fourier shapes and 2,000 mixed shapes) labelled by the fast solver (chapter 7).

- **A forward surrogate** (shape → band edges) that works for any hole family, a **multi-fidelity correction** learned from the COMSOL datasets, and **four inverse-design methods** compared honestly: every proposed shape is checked for fabrication rules and re-simulated (chapters 8–10).
- **A design gallery:** several different, verified hole shapes for one target band gap, exported in COMSOL's sweep format so you can confirm them (chapter 10).
- **The round-cell (circular-hole) data is in:** 49 COMSOL cells at k = 1, 2, 33 with a gap (up to 28.3 GHz). It joins the test set and the multi-fidelity correction (chapters 3, 6, 9).

2. Physics you need

2.1 Photonic crystals and band diagrams

A photonic crystal is a material whose refractive index repeats periodically, here silicon with air holes repeated on a lattice with period a = 242.5 µm. Light in such a structure obeys Maxwell's equations; in 2D, for the polarization used by your COMSOL model (magnetic field H_z out of the plane, electric field in the plane — called TE here), they reduce to one scalar *master equation*:

$$-\nabla \cdot \left(\frac{1}{\varepsilon(\mathbf{r})}\nabla H_z(\mathbf{r})\right) = \left(\frac{\omega}{c}\right)^2 H_z(\mathbf{r})$$

This is an eigenvalue problem: only certain frequencies ω are allowed. Because ε(r) is periodic, Bloch's theorem says the solutions are plane waves times periodic functions,

$$H_z(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\, u_{\mathbf{k}}(\mathbf{r}), \qquad u_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) = u_{\mathbf{k}}(\mathbf{r})$$

so we only need to solve inside **one unit cell** with a phase shift $e^{i\mathbf{k}\cdot\mathbf{R}}$ between opposite faces (COMSOL's *Floquet periodicity*). For each wave vector **k** we get a list of frequencies $f_1(\mathbf{k}) < f_2(\mathbf{k}) < \ldots$ — the **bands**. Plotting them along a path of k values gives the **band diagram**. If no band exists between two frequencies for *any* k, that range is a **complete band gap**:

$$f_{\text{lower}} = \max_{\mathbf{k}} f_1(\mathbf{k}), \qquad f_{\text{upper}} = \min_{\mathbf{k}} f_2(\mathbf{k}), \qquad \Delta f = f_{\text{upper}} - f_{\text{lower}}$$

These two numbers are the targets of the whole project. If Δf < 0 the bands overlap and there is no complete gap.

2.2 The lattice, the two holes and the k-path

The lattice is triangular with primitive vectors $\mathbf{a}_1 = a(1, 0)$ and $\mathbf{a}_2 = a(1/2, \sqrt{3}/2)$. Each cell holds two holes, at the two sites of a honeycomb: $A = (\mathbf{a}_1+\mathbf{a}_2)/3$ and $B = 2(\mathbf{a}_1+\mathbf{a}_2)/3$. Their distance is $a/\sqrt{3} = 140.0$ µm — which is why your L values are multiples of 140 µm. Each A hole has three nearest B neighbours at 120° from each other, and vice versa.

The reciprocal lattice has vectors $\mathbf{b}_1$, $\mathbf{b}_2$ with $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$. Its Wigner–Seitz cell, the **first Brillouin zone**, is a hexagon with special points:

$$\Gamma = (0, 0), \qquad M = \left(\frac{\pi}{a}, \frac{\pi}{\sqrt{3}a}\right), \qquad K = \left(\frac{4\pi}{3a}, 0\right)$$

Your COMSOL sweep parameter k runs along Γ → M → K → Γ, with k = 0 at Γ, 1 at M, 2 at K and 3 back at Γ (281 points, 0.1 to 2.9). Corners K and K′ of the hexagon are the two **valleys**.

2.3 Dirac cones and the valley-Hall effect

When the two holes are identical the cell has six-fold rotation symmetry (C_6), which includes inversion (r → −r). Then bands 1 and 2 touch at K and K′ in a cone-shaped crossing, a **Dirac point** — see the red dashed curves in the group's band diagram, and the dotted grey curve in figure 6.1. Near K the two bands behave like a 2×2 Dirac Hamiltonian:

$$H(\mathbf{q}) = v_D\,(q_x\sigma_x + q_y\sigma_y) + m\,\sigma_z, \qquad \Delta f \propto 2|m|$$

Making the holes different (L1 ≠ L2) breaks inversion but keeps three-fold symmetry (C_3). This adds the 'mass' m, which opens a gap at K and K′. Its sign decides the **valley Chern number** $C_K = \text{sgn}(m)/2$. Swapping which hole is big flips the sign: VPC-A and VPC-B in the group's slides are the same crystal with the holes swapped, with opposite valley Chern numbers. Where they meet, the difference in topology forces a guided **edge state** inside the gap that cannot back-scatter at sharp 120° bends — the reason VPCs are attractive for 6G chips.

Two rules that follow (they shape option B)

1. **Each hole must keep C_3 symmetry** about its own centre. A random blob breaks C_3; the Dirac points are then no longer protected at K and the gap you get is not a valley-Hall gap.
2. **The two holes must differ** (that is what breaks inversion). Swapping them gives exactly the same band edges but the opposite topology, so the dataset can be doubled for free and the design network should treat a swapped design as 'the same answer'.

2.4 Two useful invariances

Scale. For a 2D crystal of non-dispersive materials, scaling every length by s divides every frequency by s. A design found for a = 242.5 µm can be moved to another 6G channel by scaling a and the holes together.
Mirror. Reflecting the whole crystal about an A–B bond line maps K to K’; by time reversal f(K) = f(K’), so the band edges do not change. In the Fourier description this flips the sign of all sine coefficients (chapter 5).

3. The data you have: the COMSOL datasets

3.1 What COMSOL computed

The COMSOL 6.4 model is 2D: a single unit cell with Floquet-periodic boundaries, Electromagnetic Waves (emw), eigenfrequency study with 2 eigenfrequencies, swept over two size parameters and k. The two original sweeps use the same 8 × 8 grid (L1 = 140–189 µm, L2 = 91–140 µm, 7 µm steps) and the full 281-point k-path. On 28 October two more runs were added, both at k = 1 and 2 only (section 3.5):

File	Holes	k-points	Gapped cells (Δf > 0.1 GHz)	Gap range
parameter_sweep_values (... r = 2L/(3√3)).csv	rounded triangles	281	61 of 64	0.6–43.6 GHz
r1=r2=0 ; varied L1 & L2 (Parametric sweep).csv	sharp triangles	281	63 of 64	3.0–40.2 GHz
r=a_divby(3sqrt(3))_(Parametric sweep at k=1,2).csv (also split into k=1 and k=2 files)	circles (7 × 7 radii)	2	33 of 49	0.4–28.3 GHz
parameter_sweep_values (... r = 2L/(3√3)) (k=1,2).csv	rounded triangles again, finer mesh	2	61 of 64	a check of row 1

3.2 Why 29.5 million rows are only 128 designs

The export has one row **per mesh node**: X, Y, L1, L2, k, λ, emw.freq. The eigenfrequency is a single number for the whole cell, so it is repeated at all ~820 nodes. Counting carefully:

Level	Count	Meaning
rows in each file	~14.7 million	nodes × k-points × 2 eigenfrequencies × 64 geometries
unique values per file	35,968 = 64 × 281 × 2	one frequency per (geometry, k, band)
designs per file	64	one (L1, L2) each → one (f_lower, f_upper)
both files	128 designs (124 with a gap)	the training set for the old problem

src/utils/process_bands.py reduces both files to bands.csv (35,968 rows each file: geometry, k, f1, f2) and band_edges.csv (128 rows). Always split train/test *by geometry*, never by row, or the same design leaks into both sets.

3.3 The exact hole shapes (worked out for option B)

To put old and new shapes in one description we need the old outlines exactly. The dataset title says r = 2L/(3√3); the slides show the corner arcs. A normal fillet of radius r on a triangle of side L is impossible here (the largest fillet that fits is L/(2√3) = 0.29 L, but 2/(3√3) = 0.385). Measuring the schematic in the group's slides shows the corner arcs are centred on the *hole centre*: the rounded hole is the triangle **intersected with a disk**. In polar form, around the hole centre:

$$r_{\text{sharp}}(\theta) = \frac{h}{\cos\theta_f}, \qquad r_{\text{rounded}}(\theta) = \min\left(\frac{h}{\cos\theta_f},\ \rho\right), \qquad h = \frac{L}{2\sqrt{3}},\quad \rho = \frac{2L}{3\sqrt{3}} = \frac{4h}{3}$$

θ_f is the angle to the nearest flat side. The flat sides face the three neighbouring holes and the corners point into the hexagon centres, which leaves a silicon wall of at least 45 µm everywhere in the sweep. The fast solver of chapter 6 reproduces COMSOL to within ~1 GHz with exactly this geometry — a strong check that it is right. (Ask the person who built the model to confirm; the formula is in src/shapes/geometry.py.)

3.4 Where the band edges sit

In all 128 old cells band 1 peaks at K. Band 2 bottoms out at M (rounded, 63 of 64) or K (sharp, 63 of 64). So computing only k = 1 (M) and k = 2 (K) reproduces every band edge exactly — 140 times cheaper than 281 k-points. Chapter 6 shows this stays true for freeform shapes.

3.5 The round-cell dataset and the k = 1, 2 re-run

Circles. The round cell from the slides: two circular holes, r1 = 46.7–60.7 µm on site A and r2 = 32.7–46.7 µm on site B, in 7 steps of 2.33 µm each. That is r1 = 1.0–1.3 and r2 = 0.7–1.0 times a/(3√3) = 46.7 µm — the same layout as the L1/L2 triangle grid. A circle is the simplest C_3 shape (every Fourier coefficient except r_0 is zero), so it slots straight into option B. The sweep was run at k = 1 and 2 only, as section 6.4 recommends. Gap widths in GHz (negative = no complete gap):

r1 \ r2 (µm)	32.7	35.0	37.3	39.7	42.0	44.3	46.7
46.7	-2.3	-3.8	-5.3	-6.8	-8.1	-9.5	-10.7
49.0	2.0	0.4	-1.3	-2.9	-4.5	-6.1	-7.7
51.3	6.7	4.9	3.1	1.2	-0.6	-2.5	-4.3
53.7	11.6	9.7	7.7	5.7	3.6	1.5	-0.5
56.0	16.9	14.8	12.7	10.5	8.2	5.9	3.5
58.3	22.4	20.2	18.0	15.6	13.1	10.5	7.8
60.7	28.3	26.0	23.5	21.0	18.3	15.5	12.5

- The gap grows with the difference of the radii, as for the triangles, up to 28.3 GHz on this grid (the triangles reach 44 GHz).
- Equal circles (r1 = r2 = 46.7 µm) have **no** complete gap (-10.7 GHz). The two bands still touch at K (the Dirac point, 326.8 GHz), but band 2 at M is lower (316.1 GHz), so the bands overlap in frequency. A gap opens only when r1 exceeds r2 by about 9–14 µm, which is why 16 of the 49 cells have none. Circles need a larger size difference than triangles to open a gap.
- In every circle cell band 1 tops out at K and band 2 bottoms out at M — COMSOL itself agrees with the shortcut.
- The fast solver, fitted only on triangles, predicts these cells to within 0.92 / 0.39 GHz (section 6.3): the circle model uses the same permittivity as the triangle models.

Rounded triangles, re-run. The rounded sweep was repeated at k = 1 and 2 only, with a finer mesh (about 5,500 mesh-node rows per solution instead of about 820). Its band edges match the old 281-point sweep for all 64 cells to within 0.030 GHz, and the gap to within 0.012 GHz. That confirms two things at once: COMSOL's two-point route gives the same band edges as the full path, and the old mesh was already converged (to about 0.01%). It is not added as training data, because the same cells are already in the dataset.

4. Stage 1 recap: the L1/L2 benchmark

Before option B, three inverse-design methods were compared on the old data (10 random splits, errors judged by a per-family Gaussian-process 'oracle' standing in for COMSOL; src/benchmark/).

- **Direct inverse NN**: a network trained to map target → (L1, L2) directly.
- **Tandem NN** (Liu et al. 2018): first train a forward network (L1, L2) → edges, freeze it, then train the inverse network through it so the loss is measured on the *response*, not on the geometry.
- **GP + gradient**: fit a Gaussian process forward model, then solve for the geometry with multi-start least squares.

Task	Method	Rounded only	Both families, rounded test	Both families, sharp test
edges (one-to-one)	direct	0.508	0.577	0.537
	tandem	0.642	0.700	0.697
	GP + gradient	**0.007**	**0.058**	**0.060**
gap only (many-to-one)	direct	1.671 (max 30.5)	1.443	0.633
	tandem	0.165	0.170	0.090
	GP + gradient	**0.001**	**0.033**	**0.007**

Mean error of the achieved target, GHz.

Lessons that motivate option B

• With 2 design numbers and 2 targets the map is one-to-one (its Jacobian never changes sign), so there is no non-uniqueness for a tandem to solve, and 61 smooth points are best handled by a Gaussian process.
• The tandem's advantage appears as soon as the problem is many-to-one (gap-only task: 6–10× better than the direct network). Option B makes the real problem many-to-one: 10 or more shape numbers for 2 targets.
• 64 points per family cannot support freeform shapes. More data is needed — chapter 6 shows how to get it cheaply.

5. Option B: describing any C3 hole with a few numbers

5.1 The symmetry selection rule

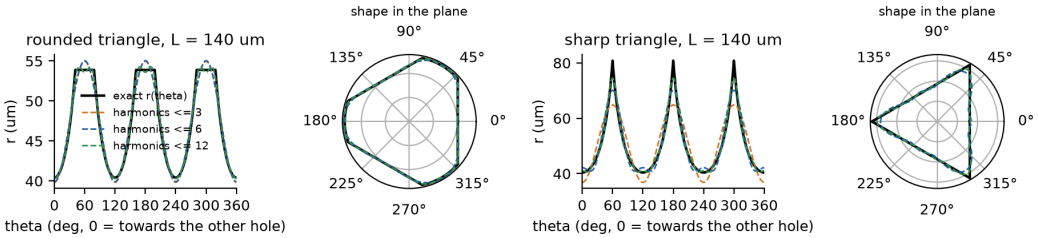
Describe a hole by its outline in polar form r(θ) around its own centre, with θ = 0 pointing at a neighbouring hole of the other type (the hole's **local frame**). Any closed star-shaped outline has a Fourier series

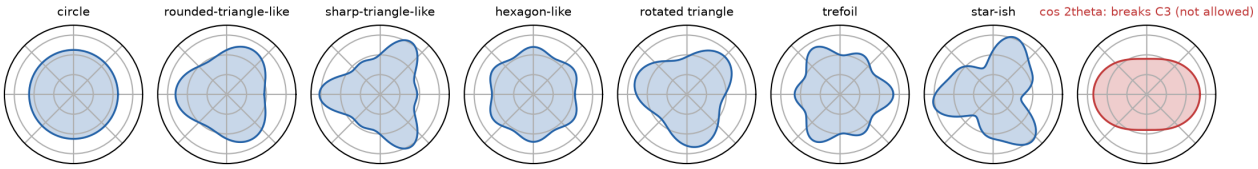
$$r(\theta) = r_0 + \sum_{n \geq 1} [c_n \cos n\theta + s_n \sin n\theta]$$

C_3 symmetry means r(θ + 120°) = r(θ). Substituting, each term picks up a factor $e^{i2\pi n/3}$, which equals 1 only when n is a multiple of 3. So **only harmonics n = 3, 6, 9, 12, ... are allowed**; a cos 2θ term, for example, would stretch the hole into an oval and break C_3:

$$r(\theta) = r_0 + c_3 \cos 3\theta + s_3 \sin 3\theta + c_6 \cos 6\theta + s_6 \sin 6\theta \ (+ c_9, s_9, c_{12}, s_{12})$$

What each number does. r_0 is the mean radius (sets the hole size and air fraction). c_3, s_3 make the hole triangular: amplitude $\sqrt{(c_3^2+s_3^2)}$, and the phase rotates it ($c_3 < 0$, $s_3 = 0$ is a triangle whose flat side faces the neighbour, like the old holes). c_6, s_6 sharpen or flatten the corners (hexagon-like, star-like). Higher harmonics add finer detail. Using (c_n, s_n) instead of (amplitude, angle) avoids the ambiguity that a triangle rotated by 60° equals a flipped triangle, so every shape has exactly one description.





Top: the exact outlines of the old rounded and sharp triangles (black) and their Fourier reconstructions with harmonics up to 3, 6 and 12. Bottom: shapes from the same formula. The red one uses cos 2ϑ and breaks C3.

5.2 One descriptor for every dataset

Each hole is turned into 9 numbers (r_0, c_3, s_3, c_6, s_6, c_9, s_9, c_{12}, s_{12}, in µm), computed numerically from the exact outline. A unit cell is 18 numbers (hole A then hole B). The old shapes become:

Old hole	r0 / L	c3 / r0	c6 / r0	c9 / r0	c12 / r0
rounded triangle (∩ disk)	0.339	−0.160	0.002	0.019	−0.011
sharp triangle	0.363	−0.275	0.104	−0.054	0.032
circle, radius R	R / R = 1	0	0	0	0

So the 'corner type' flag of stage 1 is no longer needed: a sharp triangle is simply a hole with a stronger 3rd harmonic and a non-zero 6th harmonic. A surrogate trained on these 18 numbers can, in principle, learn from rounded, sharp, circular and freeform holes at once, and a circle-hole COMSOL sweep adds directly to it.

5.3 The design space for inverse design

For designing new holes we use the first two harmonics for each hole: **10 numbers per cell** $[r_0, c_3, s_3, c_6, s_6]_A + [...]_B$. They are sampled in a box that contains the old triangles: r_0 in [25, 75] µm, $\sqrt{(c_3^2+s_3^2)} \le 0.35\, r_0$, $\sqrt{(c_6^2+s_6^2)} \le 0.12\, r_0$, phases uniform. The networks work with u in $(0, 1)^{10}$ ($r_0 = 25 + 50u$, $c_3 = 0.35\, r_0(2u - 1)$, ...), so a sigmoid output is always a shape inside the box. A second 'library' design space lets each hole be a circle, a rounded or sharp triangle (any size and rotation) or a Fourier shape — the 'mix of 3–4 shapes'.

5.4 Fabrication and validity rules

Random coefficients can produce cells that cannot be made, or that are not a valley crystal at all. Every cell is checked (geometry.check_cell) before it is simulated or proposed:

- silicon wall between any two holes (A–B neighbours, A–A and B–B second neighbours, periodic images) **≥ 25 µm** — the old data never goes below 45 µm;
- hole radius everywhere **≥ 10 µm** (no slivers; also guarantees a simple, non-self-intersecting outline);
- radius of curvature of convex corners **≥ 8 µm** (deep reactive-ion etching cannot cut sharper corners);
- the holes do not overlap, and each stays C_3 by construction.

About 40–60% of random draws pass. These limits are parameters (check_cell(min_wall=..., ...)); set them to your fabrication process. For inverse design the same rules are written as a smooth penalty P(u) (models.penalty) that is zero only for comfortably valid cells, so gradients can push a design back into the valid region.

5.5 Different designs that are physically the same

Two operations never change the band edges: swapping hole A and hole B (chapter 2.3) and mirroring the cell (all sine coefficients change sign). When we count 'how many different shapes reach a target' we treat these as the same design, so the diversity numbers in chapter 10 are honest.

6. A fast simulator: plane-wave expansion (PWE)

6.1 Why

Freeform shapes need thousands of labelled examples; 128 COMSOL runs are far too few, and thousands of COMSOL runs take days. Your model is 2D, and for a 2D photonic crystal the classic plane-wave expansion method is fast and accurate. It is used here as a **cheap, slightly less accurate simulator**: it labels thousands of shapes, the machine learning is trained on them, and only the final designs go back to COMSOL.

6.2 How it works

Expand the periodic part of the field in plane waves $e^{i(k+G)\cdot r}$ over reciprocal lattice vectors $G = m\,b_1 + n\,b_2$, and expand $1/\varepsilon$ the same way. The master equation becomes a matrix eigenproblem:

$$\sum_{\mathbf{G}'} (\mathbf{k}+\mathbf{G})\cdot(\mathbf{k}+\mathbf{G}')\;[\varepsilon^{-1}]_{\mathbf{G}\mathbf{G}'}\;h_{\mathbf{G}'} = \left(\tfrac{\omega}{c}\right)^2 h_{\mathbf{G}}$$

The permittivity map of the cell is drawn on a 96 × 96 grid with 4 × 4 sub-pixel averaging (so smooth shape changes give smooth frequency changes) and Fourier-transformed with an FFT:

$$\varepsilon_{\mathbf{G}} = \frac{1}{A_{\text{cell}}}\int_{\text{cell}} \varepsilon(\mathbf{r})\, e^{-i\mathbf{G}\cdot\mathbf{r}}\, d^2r$$

$[\varepsilon^{-1}]$ is the inverse of the matrix $\varepsilon_{G-G'}$ (Ho, Chan and Soukoulis 1990), which converges much faster than transforming $1/\varepsilon$ directly. With 439 plane waves the band edges change by less than 0.4 GHz when going to 1,251 plane waves. One cell takes ≈ 0.05 s for the matrix plus ≈ 0.06 s per k-point; with M and K only, about 0.17 s on one CPU core, and the generator runs 10 cells in parallel. Code: src/shapes/pwe.py.

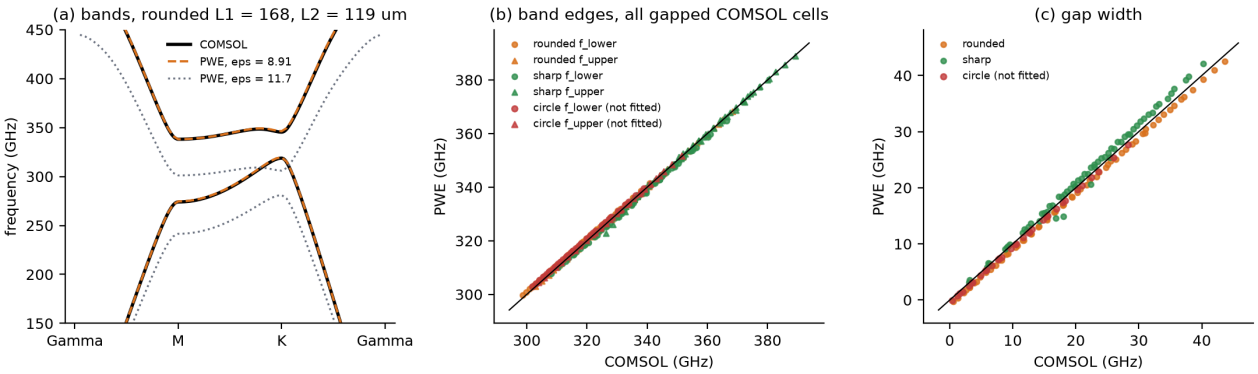
6.3 Checking it against the COMSOL datasets

With the slides' value ε = 11.7 the solver is about 13% too low at every k and for both shape families — the same ratio everywhere, which is the signature of a different permittivity, not a wrong geometry. Fitting ε on every 4th geometry gives **ε = 8.914 (n = 2.986)**; then all 128 cells are reproduced:

Family	cells	MAE f_lower	MAE f_upper	MAE gap	gap correlation
rounded	61	0.88	0.17	0.98	0.99993
sharp	63	0.87	0.43	1.01	0.9970
circle (not used to fit ε)	33	0.92	0.39	0.53	0.99999

Errors in GHz (about 0.1–0.3% of the frequency).

The circle row is the real test. ε was fitted on triangles only, yet the 33 gapped circle cells come out within 1.03 GHz (f_{lower}) and 0.53 GHz (f_{upper}) at worst. PWE is slightly high for circles (+0.92 / +0.39 GHz on average), so its gaps are 0.53 GHz too narrow. It agrees with COMSOL on which cells have a gap in 48 of 49 cells; the exception has a gap close to zero.



(a) Full band diagram: COMSOL (black), PWE with the fitted ε (orange dashes, on top of black) and with ε = 11.7 (grey). (b, c) Band edges and gap widths of all 128 triangle cells, plus the gapped circle cells (red), which were not used to fit ε.

What the ε = 8.9 result means — please check

A 2D model that should mimic a 200 µm silicon slab often uses the slab's *effective index* instead of bulk silicon. n ≈ 2.99 is a plausible effective index for this slab near 300 GHz. Open the COMSOL model → Materials (or the emw node) and check the relative permittivity / refractive index. If it is 11.7, something else differs (for example an out-of-plane wave number), and the fitted ε simply absorbs it — the comparison above shows the PWE reproduces the model either way. In the thesis, describe it as 'calibrated to COMSOL'.

Published support: Shalaev et al. (Nature Nanotechnology 2019) reduce a silicon membrane to a 2D COMSOL model in exactly this way and use n_{eff} = 2.965 — close to the n = 2.986 fitted here. Cite them when you explain the 2D model.

6.4 Why COMSOL needs only k = 1 (M) and k = 2 (K)

The band edges are defined over *all* wavevectors: f_{lower} is the highest point of band 1 and f_{upper} the lowest point of band 2 anywhere in the Brillouin zone. Your COMSOL sweep samples 281 points along the path Γ → M → K → Γ (k = 0 → 1 → 2 → 3). The shortcut used everywhere in this project goes further and solves only at M and K, about 140 times less work. This section explains why that is justified, what the literature does and does not promise, and how it was checked.

What the literature says

- **Why K.** With two identical holes the honeycomb crystal has six-fold symmetry and bands 1 and 2 touch at K (a Dirac point). Making the holes different breaks inversion symmetry, the touching point splits and the gap opens *at K*, so the top of band 1 sits at K. This is the standard valley-Hall picture (Ma and Shvets 2016; Shalaev et al. 2019; Yang et al. 2020).
- **Why M.** Band 2 is not necessarily lowest at K: away from K it can dip, and its lowest point can be M instead. In your COMSOL data it is M for the rounded triangles (63 of 64 cells) and K for the sharp ones (63 of 64). Shalaev et al. (2019) chose triangular holes precisely because they push band 2 up at M. With circular holes in their silicon slab, band 2 dipped between K and M and the gap became smaller and indirect — the case a two-point shortcut would miss.
- **No guarantee.** Band-structure codes usually scan only the edge of the irreducible Brillouin zone, but that is a rule of thumb, not a theorem. Harrison, Kuchment, Sobolev and Winn (2007) built periodic structures whose band edges lie deep inside the zone. Craster et al. (2012) warn of the same for waves in periodic media. Maurin et al. (2018) ran hundreds of simulations covering all 17 two-dimensional symmetry groups and found the edge-only search reliable for highly symmetric lattices but less so for low-symmetry ones.
- **Where your crystals stand.** The old triangle cells have three mirror lines, and with time reversal their irreducible zone is the triangle Γ–M–K: the COMSOL path is exactly its boundary. A freeform hole with sine terms has no mirror line (plane group p3, rotations only — the low-symmetry case). Its irreducible zone is the whole 60° wedge Γ–K–K′, which contains points the COMSOL path never visits. So the shortcut

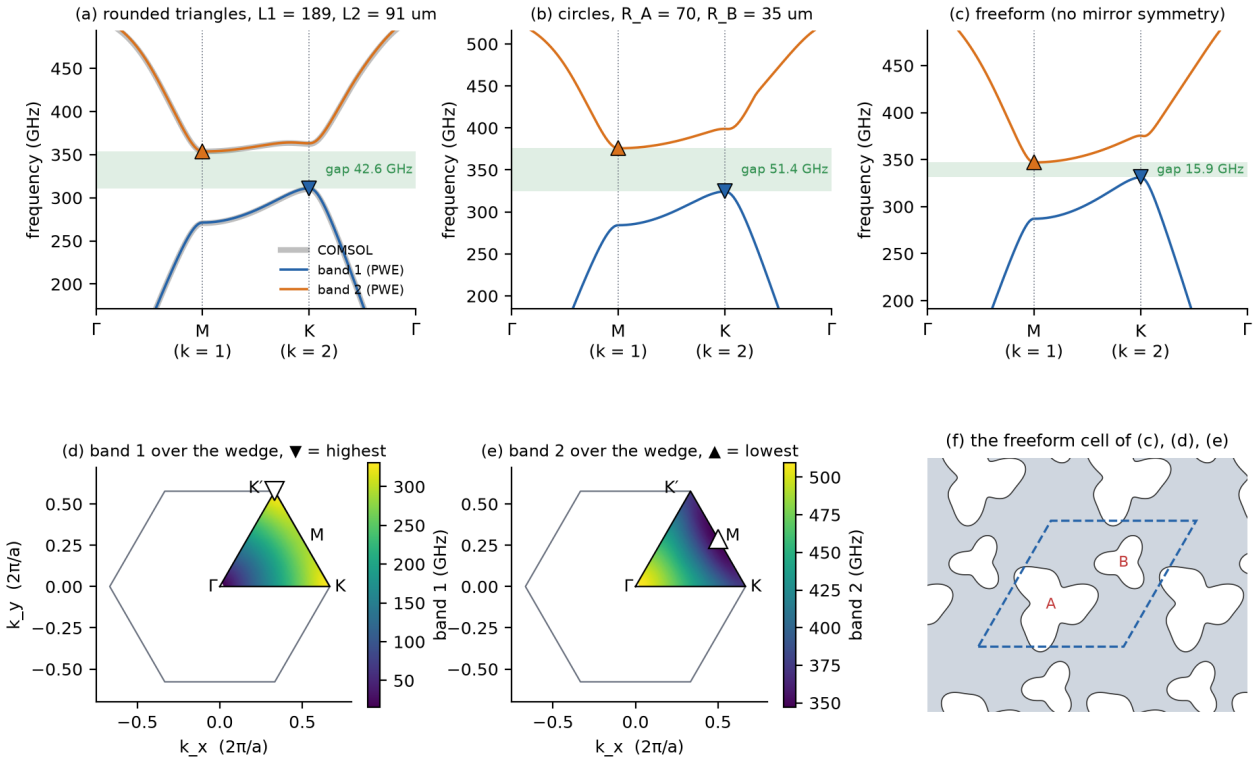
has to be *checked*, especially for freeform shapes and for circles.

Three checks

Each check compares the band edges from M and K alone with the band edges from a much denser set of wavevectors. The script is src/shapes/check_mk_shortcut.py; the numbers are in results/optionb/mk_check.json.

Check	Cells	Compared against	Largest difference	Top of band 1 at	Bottom of band 2 at
COMSOL: rounded and sharp triangles	128	the full 281-point COMSOL path	0 (exact)	K (128)	M (64), K (64)
PWE: random mixed cells (freeform, rotated triangles, circles)	658	29-point path + 27-point grid over the wedge	0.003 GHz	K (658)	M (613), K (45)
PWE: circle pairs, radii 20–95 µm	80	149-point path + 65-point wedge grid	< 0.001 GHz	K (80)	M (80)
COMSOL: rounded triangles re-run at k = 1, 2 only (finer mesh)	64	the old 281-point sweep	0.030 GHz (mesh)	–	–

In total 738 PWE cells (106 of them circle pairs) and all 128 COMSOL cells: no band edge lies anywhere other than M or K.



(a–c) Bands along the COMSOL path for three kinds of cell. Downward triangle = highest point of band 1, upward triangle = lowest point of band 2, green = the gap. In (a) the thick grey line underneath is COMSOL itself. (d, e) The same two bands of the freeform cell over the whole wedge Γ–K–K′ (black triangle; grey hexagon = Brillouin zone). The highest point of band 1 is a K corner (K and K′ give the same frequency) and the lowest point of band 2 is M, the middle of the K–K′ edge — nowhere inside the wedge.

How to read the figure. Along the path (top row) band 1 rises to its maximum at K and, in these three cells, band 2 falls to its minimum at M (for sharp triangles it is usually K; either way a point the shortcut solves). The bottom row answers the question the path cannot: could a band edge hide *off* the path, inside the wedge? The colour maps show it does not — band 1 rises steadily towards the corners K and K′, and band 2 is lowest at M on the zone edge.

For your round-cell (circle) data

All 80 circle pairs tested (radii 20–95 µm, walls ≥ 20 µm) have the top of band 1 at K and the bottom of band 2 at M, with a largest difference of < 0.001 GHz. The dip between K and M that Shalaev et al. saw for circles in a 3D slab does not appear in your 2D model.

27 of these pairs (nearly equal radii) have no complete gap at all: band 2 at M falls below band 1 at K. The two-point shortcut still reports this correctly, as a negative gap.

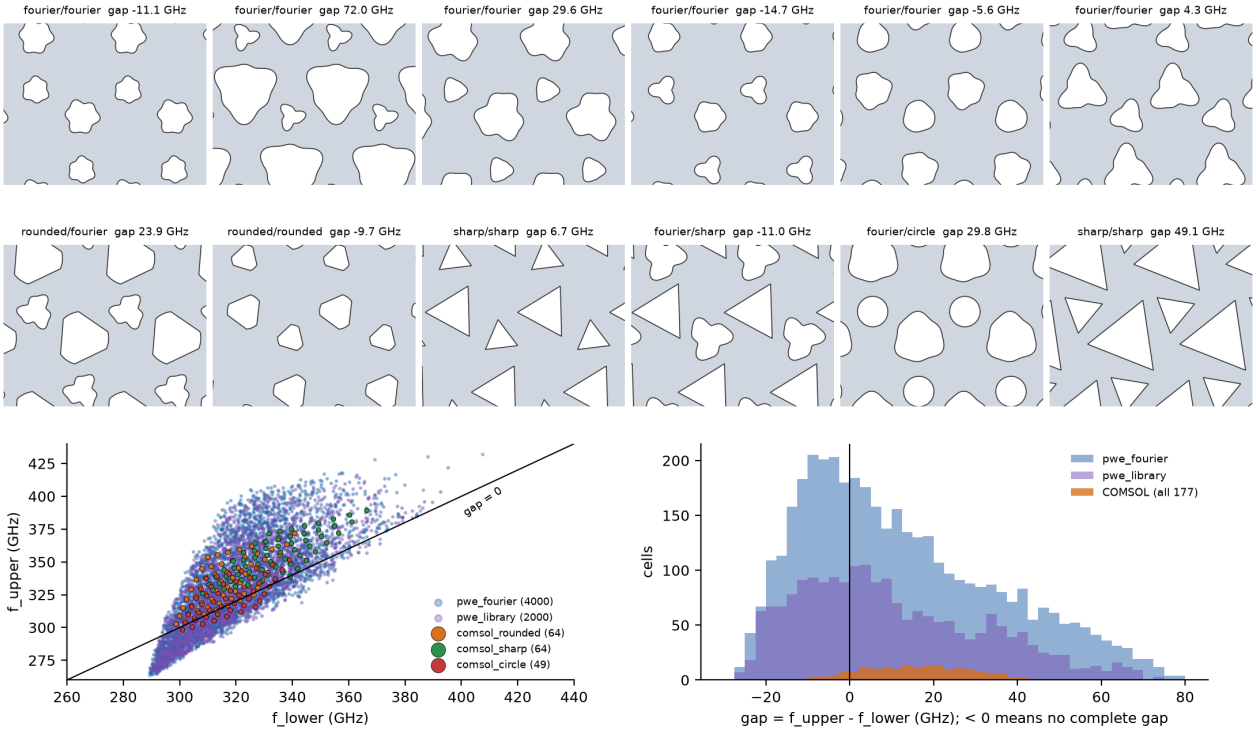
Your COMSOL circle sweep (section 3.5, k = 1 and 2 only) agrees with the fast solver at M and K to within about 1 GHz, and in COMSOL too band 1 tops out at K and band 2 bottoms out at M in every cell. The one direct COMSOL check still missing is a full-path run of 2–3 circle cells (a few minutes); do it once for the thesis.

How to write it in the thesis

“Band extrema of two-dimensional photonic crystals usually, but not always, lie on the boundary of the irreducible Brillouin zone [Harrison 2007; Craster 2012; Maurin 2018]. We therefore verified this for our structures. For all 128 COMSOL cells, the band edges at M and K equal those over the full 281-point path. For 738 plane-wave cells, including 106 with circular holes, they equal those over a dense path and over the full irreducible zone to within 0.003 GHz. The gap opens at K through inversion-symmetry breaking, and the second band near M limits the upper edge [Shalaev 2019].”

7. The option-B dataset

src/shapes/gen_dataset.py draws random cells, keeps the valid ones and labels them with the PWE solver at M and K (10 processes in parallel). Two sets were generated: **fourier** (4,000 cells, both holes Fourier shapes) and **library** (2,000 cells, each hole independently a circle, rounded triangle, sharp triangle or Fourier shape, with random size and rotation). Each row stores the two hole families and parameters, the 18 Fourier features, f_{lower}, f_{upper}, the gap, where each edge sits (M or K), the thinnest wall and the air fraction.



Top: random cells from the fourier (first row) and library (second row) sets. Bottom left: every cell as a point (f_lower, f_upper); cells above the diagonal have a complete gap. The COMSOL cells (orange, green) sit inside the new cloud. Bottom right: gap distribution.

Two things to notice. First, freeform shapes reach much wider gaps than the old triangles (about 70 GHz versus 44 GHz) and a wider range of centre frequencies, which is the point of a larger design space. Second, a sizeable fraction of cells have no complete gap ($\Delta f < 0$). They are kept for training: they teach the surrogate where gaps close, which the inverse methods need to know.

8. Forward surrogate: shape → band edges

8.1 The model

Input: the 18 Fourier features of the cell (µm), standardised. Output: (f_{lower}, f_{upper}) in GHz, standardised. Each network is a multilayer perceptron with three hidden layers of 256 SiLU units, trained with AdamW and a one-cycle learning-rate schedule (400 epochs, batch 128) on 90% of the 6,000 PWE cells — *including* cells without a gap, because they teach the model where gaps close. Five networks with different random seeds form a **deep ensemble**:

$$\hat{\mathbf{y}}(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} F_m(\mathbf{x}), \qquad \hat{\sigma}(\mathbf{x}) = \mathrm{std}_m\, F_m(\mathbf{x}), \qquad M = 5$$

The mean is the prediction; the spread is a cheap uncertainty estimate (large where the members disagree, i.e. far from the data). The baseline is a Gaussian process with an automatic-relevance (one length scale per feature) kernel:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{1}{2}\sum_{i=1}^{18}\frac{(x_i - x'_i)^2}{\ell_i^2}\right) + \sigma_n^2\,\delta_{\mathbf{x}\mathbf{x}'}$$

8.2 How many simulations are needed?

The learning curve answers the most practical question of the project: how many labelled cells (PWE now, COMSOL later) does a freeform surrogate need? Same held-out test cells for every row; mean absolute error in GHz for (f_{lower}, f_{upper}).

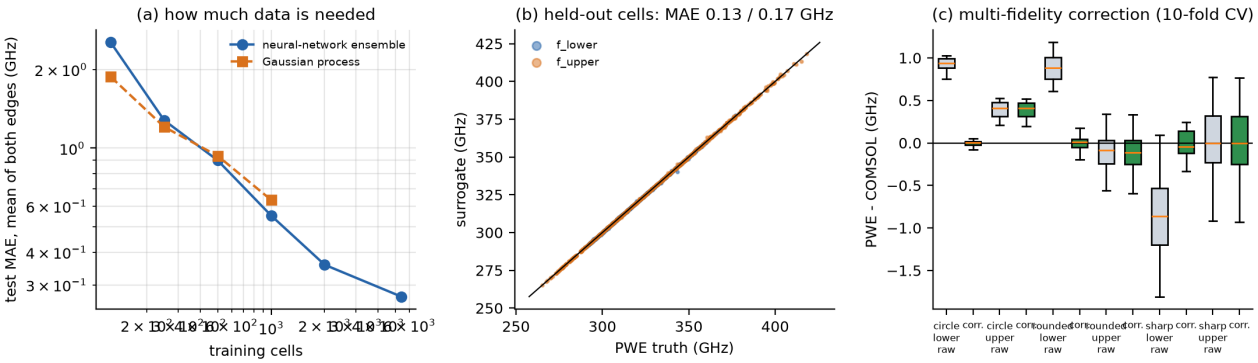
training cells	NN ensemble MAE	GP MAE	NN time (s)	GP time (s)
125	2.48 / 2.59	2.44 / 1.31	23.6	1.6
250	1.22 / 1.33	1.48 / 0.94	43.4	6.6
500	0.82 / 0.97	1.17 / 0.70	61.3	34.5
1000	0.46 / 0.64	0.73 / 0.54	61.3	147.9
2000	0.30 / 0.41	— (too slow)	57.2	—
5400	0.23 / 0.31	— (too slow)	55.5	—

With 125 cells the GP is about as good as or better than the network (1.87 vs 2.54 GHz on average) — the stage-1 lesson again: small data favours GPs. With more data the network keeps improving (at 1000 cells: NN 0.55 vs GP 0.63 GHz; with all 5,400 cells the network reaches 0.27 GHz), while an exact GP grows as n^3 in cost. **Planning rule:** about 500 COMSOL cells already give ~1 GHz accuracy over this whole freeform space. With the 2-k-point shortcut that is 1,000 eigen-solves — time your 3-geometry check run to know whether that is an evening or a weekend.

8.3 Accuracy of the final ensemble

test set	MAE f_lower	MAE f_upper	MAE gap	max error
all 600 held-out PWE cells	0.13	0.17	0.21	3.1 / 2.1
only cells with a gap	0.12	0.22		
pwe_fourier	0.12	0.16		
pwe_library	0.16	0.20		

GHz. Correlation between the ensemble spread and the actual error on the test set: 0.61 (positive = the uncertainty is informative).



(a) Learning curve, log–log. (b) Surrogate vs PWE on held-out cells. (c) Error of the PWE solver against COMSOL before (grey) and after (green) the multi-fidelity correction, per dataset and band edge.

Note the last row. A network trained *only* on PWE labels of random cells predicts the COMSOL cells within about a gigahertz. Its training data contained triangles of random size and rotation, but none of these exact cells, and no COMSOL labels at all. Part of that error comes from the network and part is the PWE–COMSOL difference, which chapter 9 corrects.

9. Using the COMSOL datasets: multi-fidelity correction

We have two simulators: PWE (fast, slightly biased) and COMSOL (slow, the reference). The standard trick (Kennedy and O'Hagan 2000) is to model the high-fidelity answer as the low-fidelity answer plus a smooth correction learned from the few cells where both exist. The correction δ is a Gaussian process fitted to the differences $f_{COMSOL}(x_i) - f_{PWE}(x_i)$ at the COMSOL cells, using the same 18 features:

$$f_{\mathrm{COMSOL}}(\mathbf{x}) \;\approx\; f_{\mathrm{PWE}}(\mathbf{x}) + \delta(\mathbf{x}), \qquad \delta(\mathbf{x}) \sim \mathcal{GP}$$

All COMSOL datasets are used here (circle 850, freeform 299, freeform (AI designs) 14, rounded triangle 1055, sharp triangle 1063; 3281 gapped cells). This is where using several datasets matters most: one family alone cannot tell whether the PWE bias depends on the shape. Results (10-fold cross-validation, so every number is on cells the correction did not see):

predictor	MAE f_lower	MAE f_upper	MAE gap
PWE alone (ε fitted)	0.79	0.34	
PWE + GP correction	0.14	0.23	

Does the correction transfer to a family it has not seen?

held-out family	PWE alone	PWE + correction learned on the other families
circle	0.95 / 0.40	0.04 / 0.25
rounded triangle	0.92 / 0.18	0.11 / 0.27
sharp triangle	0.63 / 0.40	1.12 / 0.38
freeform	0.44 / 0.53	0.24 / 0.67

MAE (f_{lower} / f_{upper}), GHz. Learned on the other families, the correction helps for circle, rounded triangle, freeform but **not** for sharp triangle: part of the bias is shared, part is shape-specific. This is the honest reason to add COMSOL cells of each new shape family — now the gallery designs — before trusting corrected predictions for freeform shapes.

9.1 Freeform shapes in COMSOL

The hole outline was rebuilt in COMSOL from the Fourier formula (Parametric Curve + Convert to Solid, with the Air material following the holes; data/comsol/new/FOURIER_COMSOL_STEPS.txt) and random freeform cells were run at k = 1 and 2, exported as Global Evaluation tables. So far: **1007 valid freeform cells** (299 with a gap). Two plain-circle cells in batch 1 reproduced the earlier circle sweep within 0.013 GHz, so the Fourier-hole model is set up correctly.

The PWE solver is accurate for freeform shapes too: PWE − COMSOL = 0.45 / 0.43 GHz on average (f_{lower} / f_{upper}), largest 1.38 / 1.45 GHz — the same accuracy as for triangles and circles, with no refitting. This was the largest open question of option B.

A limitation of the COMSOL cell

In the rhombic COMSOL unit cell each hole centre is only 70 µm from two cell edges. A hole that reaches further is cut off at the edge in COMSOL (the part outside is not wrapped around to the other side), so COMSOL simulates a different shape. In batch 1, 105 of 202 random cells did this and disagreed with PWE by up to 88 GHz; they are removed automatically (src/comsol/import_global_eval.py) and later batches only contain holes reaching at most 68 µm. PWE has no such limit. To allow larger holes in COMSOL, the holes would have to be copied by the lattice vectors and clipped to the cell.

The first COMSOL check of AI-designed shapes

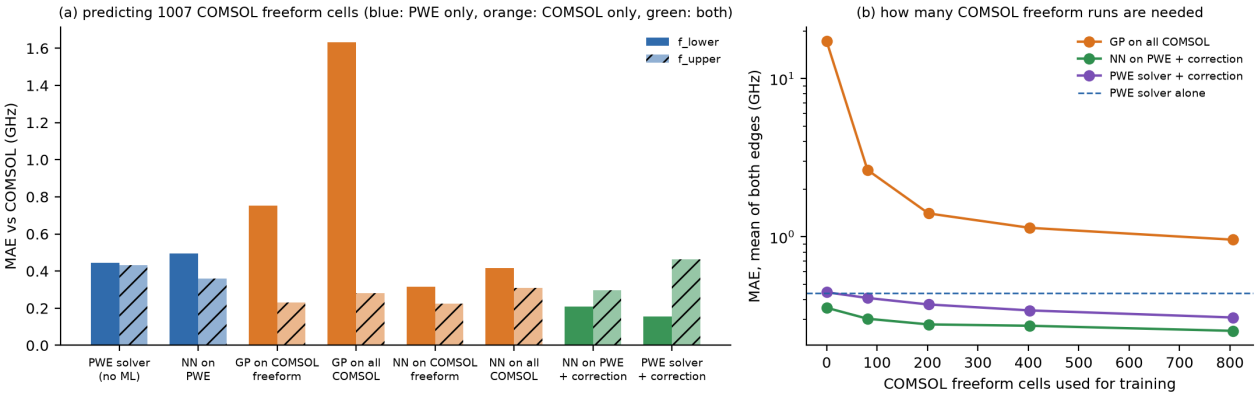
gallery design	target (GHz)	PWE (GHz)	COMSOL (GHz)	COMSOL – target
1	300 / 330	300.11 / 330.15	299.10 / 330.25	-0.90 / +0.25
2	300 / 330	299.99 / 330.00	299.11 / 330.06	-0.89 / +0.06

Shapes proposed by the inverse methods for a 300 / 330 GHz gap and simulated in COMSOL hit the target within about 1 GHz. 23 of the 24 gallery designs fit inside the COMSOL cell and can be checked the same way.

Do we still need PWE? Training on PWE vs on COMSOL

Every predictor below is scored on the 1007 COMSOL freeform cells with 5-fold cross-validation (each cell is predicted by models that never saw it; 3 repetitions). 'All COMSOL cells' adds the 3191 triangle and circle cells to the training folds. GHz, src/shapes/compare_training.py:

predictor	COMSOL data used	MAE f_lower	MAE f_upper	MAE gap
PWE solver, no ML	none	0.45	0.43	0.74
NN trained on 6,000 PWE cells	none	0.49	0.36	0.73
GP on COMSOL freeform cells only	yes	0.75	0.23	0.76
GP on all COMSOL cells	yes	1.63	0.28	1.66
nn_comsol_free	yes	0.32	0.23	0.35
NN on all COMSOL cells	yes	0.42	0.31	0.48
PWE-trained NN + COMSOL correction	yes	0.21	0.30	0.36
PWE solver + COMSOL correction	yes	0.15	0.46	0.47



(a) Each predictor's error on held-out COMSOL freeform cells. (b) Error vs the number of COMSOL freeform cells used for training (the triangle and circle COMSOL cells are always included).

Best: PWE-trained NN + COMSOL correction (0.25 GHz averaged over both edges). PWE alone gives 0.43 GHz; adding the COMSOL correction lowers that to 0.25 GHz. Here a COMSOL-only model is competitive (best 0.27 GHz vs 0.25 GHz with PWE + correction), so with this much COMSOL data the PWE stage matters less.

The learning curve (b) holds a surprise: the correction already reaches 0.36 GHz with **zero** freeform COMSOL cells, because the PWE-vs-COMSOL offset learned on the triangle and circle cells carries over to freeform shapes. The GP on COMSOL alone needs every freeform cell to get down to 1.6 / 0.3 GHz. Adding the triangle cells even hurts it (0.96 vs 0.49 GHz), because triangles say little about freeform shapes. More freeform batches (2–5) will mainly help the COMSOL-only line; rerun compare_training.py after each import to see it.

10. Inverse design: from target band edges to shapes

10.1 The problem

Given a target t = (f_{lower}, f_{upper}), find designs u in $(0, 1)^{10}$ (chapter 5.3; ϕ(u) turns u into the 18 features) that the surrogate F maps onto t while staying fabricable. All four methods approximately solve

$$\mathbf{u}^\star = \arg\min_{\mathbf{u}\in(0,1)^{10}} \sum_{j=1}^{2}\left(\frac{F_j(\phi(\mathbf{u})) - t_j}{s_j}\right)^2 + \lambda\, P(\mathbf{u})$$

where s_j is the standard deviation of each target in the data and P the validity penalty. With 10 unknowns and 2 equations there is typically an 8-dimensional family of solutions — this is the non-uniqueness that makes plain 'target → design' regression fail (it averages different valid answers into an invalid one).

10.2 Method 1 — tandem network

An inverse network G_θ: t → u (two inputs, three hidden layers of 256, ten sigmoid outputs) is trained *through the frozen surrogate* on the targets of the training cells:

$$\mathcal{L}_{\text{tandem}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\left\|\frac{F(\phi(G_\theta(\mathbf{t}_i))) - \mathbf{t}_i}{\mathbf{s}}\right\|^2 + \lambda\, P(G_\theta(\mathbf{t}_i))\right]$$

Because the loss is measured on the response, the network may pick any one of the valid designs and is never penalised for not matching 'the' training geometry — Liu et al.'s fix for non-uniqueness. Limitation: it is a function, so it returns exactly one design per target.

10.3 Method 2 — conditional variational autoencoder (CVAE)

A generative model learns the whole *distribution* of designs that produce a target. An encoder q(z | u, t) compresses a training design into a 4-number latent code z; a decoder D(z, t) rebuilds the design from the code and the target. Training maximises the evidence lower bound:

$$\mathcal{L}_{\text{CVAE}} = \mathbb{E}_{q(\mathbf{z}|\mathbf{u},\,\mathbf{t})}\left[\frac{\|\mathbf{u} - D(\mathbf{z},\mathbf{t})\|^2}{\sigma^2}\right] + \beta\,\mathrm{KL}\big(q(\mathbf{z}|\mathbf{u},\mathbf{t})\,\|\,\mathcal{N}(\mathbf{0}, I)\big)$$

At design time the encoder is dropped: draw $z^{(1)}$, ..., $z^{(64)}$ from N(0, I) and decode each with the target — 64 different candidate shapes in one pass. The CVAE is trained on real (design, response) pairs only, so its samples look like designs from the dataset; it does not use the surrogate.

10.4 Method 3 — multi-start gradient search

No inverse network at all: start from 64 random designs, and run 300 Adam steps on the objective of 10.1 directly, back-propagating through the frozen surrogate (the design is parametrised by logit(u) so it stays inside the box). Different starts end in different solutions. This is the neural-network version of stage 1's 'GP + gradient', the method that won there.

10.5 Method 4 — CVAE + refine

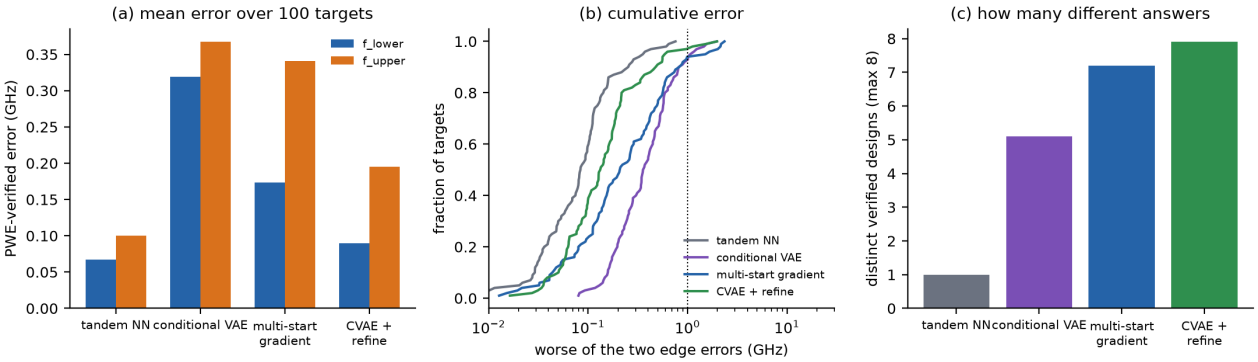
Use the 64 CVAE samples as starting points for 100 gradient steps: the CVAE supplies realistic, diverse starts and the gradient makes them hit the target precisely.

10.6 How the methods were judged

100 targets were taken from **held-out** Fourier cells with a gap above 2 GHz (so every target is reachable and no method saw it). For each method and target the candidates are ranked by surrogate error; the best one that passes the **exact** fabrication check is kept and **re-simulated with the PWE solver**. The error reported is |PWE − target|, never the surrogate's own opinion. 'Distinct designs' counts, for 10 targets, how many of the

8 best different candidates (after removing A↔B-swapped and mirrored duplicates) are verified within 1 GHz.

method	valid design found	verified MAE f_lower / f_upper	median	within 1 GHz	distinct designs (of 8)	ms per target
Tandem NN	100/100	0.07 / 0.10	0.05 / 0.07	100%	1 (by design)	0
Conditional VAE	100/100	0.32 / 0.37	0.24 / 0.29	92%	5.1	0
Multi-start gradient	100/100	0.17 / 0.34	0.11 / 0.16	94%	7.2	591
CVAE + refine	100/100	0.09 / 0.20	0.06 / 0.10	97%	7.9	200



(a) PWE-verified error per method. (b) Fraction of targets reached within a given error. (c) Number of different verified shapes per target.

Repeatability: 5 different random splits

Every number above comes from one random choice of training and test cells and one set of initial network weights. The whole experiment was repeated with 5 different random seeds (seeds [0, 1, 2, 3, 4]; src/shapes/summarize_seeds.py). Mean ± standard deviation over the runs — use these in the thesis:

method	within 1 GHz (%)	MAE f_lower (GHz)	MAE f_upper (GHz)	distinct designs (of 8)
Tandem NN	99 ± 1	0.08 ± 0.01	0.12 ± 0.03	1 (by design)
Conditional VAE	93 ± 2	0.33 ± 0.01	0.36 ± 0.05	5.1 ± 0.4
Multi-start gradient	92 ± 3	0.18 ± 0.04	0.36 ± 0.03	7.5 ± 0.3
CVAE + refine	97 ± 1	0.09 ± 0.02	0.20 ± 0.04	8.0 ± 0.1

Forward surrogate MAE on held-out PWE cells: 0.14 ± 0.01 / 0.18 ± 0.01 GHz (f_lower / f_upper). Multi-fidelity, PWE + GP vs COMSOL (10-fold CV): 0.09 ± 0.00 / 0.32 ± 0.00 GHz.

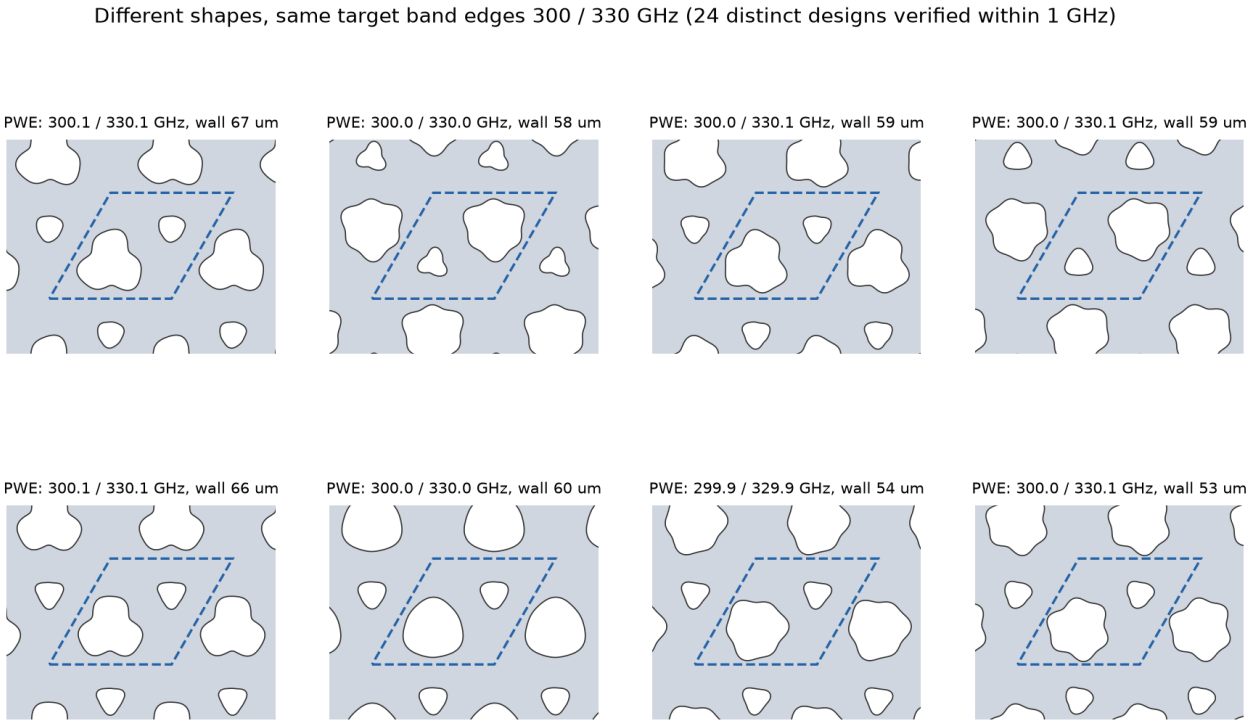
10.7 What the comparison shows

- **Most accurate: Tandem NN** (100% of targets within 1 GHz after physics verification; mean error 0.07 / 0.10 GHz). Now that the problem is many-to-one, the base-paper method does what it was designed for — compare stage 1, where it was the worst method on the one-to-one task.
- **Why does the per-target search lose to the tandem?** Multi-start optimisation has a lower *surrogate* error but a higher *verified* error (0.17 / 0.34 GHz). Searching hard against a surrogate finds the places where the surrogate is wrong in your favour (often near the edge of the data). The tandem network learns one smooth map from all training targets, which keeps its designs where the surrogate is reliable — an implicit regulariser. This 'surrogate exploitation' effect is worth a slide; the remedy is to penalise surrogate uncertainty (ensemble spread) during the search.
- **Best overall trade-off: CVAE + refine** (97% within 1 GHz, 7.9 of 8 candidates are different verified designs). The CVAE proposes realistic shapes from the data distribution, so the short refinement does not wander into unreliable regions.
- **Most different answers: CVAE + refine** (7.9 of 8). The tandem returns exactly one design per target. Having several gives a real choice: thickest walls, smoothest corners, or the best fabrication tolerance.

- **The plain CVAE** reaches 92% within 1 GHz: its samples are diverse but only approximately on target. That is why the 'refine' step helps.
- **Speed:** the tandem needs 0.1 ms per target (one forward pass); multi-start 591 ms. COMSOL would take minutes per evaluation.
- All errors here are against PWE. Against COMSOL, add the PWE–COMSOL difference from chapter 9 (about 0.3–1 GHz) until the gallery designs are checked in COMSOL.

10.8 The design gallery: many shapes, one target

Target f_{lower} = 300 GHz, f_{upper} = 330 GHz (a 30 GHz gap overlapping the upper half of the IEEE 802.15.3d band). 1024 candidates were generated by the CVAE, multi-start search and CVAE + refine; after the exact fabrication check, duplicate removal and PWE verification, **24 genuinely different designs hit the target within 1 GHz**:



Verified designs for the same target. Each panel: tiled unit cells; the dashed rhombus is one cell. The title gives the PWE-verified band edges and the thinnest silicon wall.

#	hole A: r0, c3, s3, c6, s6 (µm)	hole B: r0, c3, s3, c6, s6 (µm)	PWE edges (GHz)	wall
0	57.7, -11.4, 1.3, -2.6, -1.9	25.7, -2.0, 0.6, 0.2, -0.4	300.1 / 330.1	67
1	26.0, -4.7, 3.9, 0.7, -0.1	58.3, -4.5, -0.6, 4.0, 1.1	300.0 / 330.0	58
2	58.1, -6.3, 4.7, 2.5, 3.3	26.1, -3.4, 1.0, 0.7, 0.2	300.0 / 330.1	59
3	26.3, -3.2, -0.1, 0.4, 0.1	58.6, -4.2, 0.4, 0.7, 3.8	300.0 / 330.1	59
4	57.7, -10.9, 0.4, -3.4, 0.5	25.9, -4.1, 0.3, 0.7, 0.0	300.1 / 330.1	66
5	58.9, -3.7, 0.1, 0.8, -0.2	26.5, -3.1, 0.0, 0.3, 0.2	300.0 / 330.0	60
6	58.9, 2.9, -3.2, -2.8, -1.3	25.6, -4.0, 1.8, 0.3, -0.4	299.9 / 329.9	54
7	58.7, 0.1, 3.4, 2.9, 3.7	25.8, -2.5, 2.1, -0.4, -0.2	300.0 / 330.1	53
8	25.4, -0.0, -2.1, -0.5, 0.2	58.0, -4.1, -8.5, 3.8, -0.3	300.0 / 330.0	56
9	58.4, -4.0, -1.2, 3.2, 2.8	26.0, -5.8, 0.1, -0.8, -0.7	299.9 / 330.1	59

These numbers are exactly what goes into the COMSOL Parametric Curve of chapter 11.2 (data/pwe/showcase_for_comsol.txt has them in metres, ready for Load from File).

Read the table like a physicist. Every design pairs a large hole (median $r_0 \approx 58$ µm) with a small one (≈ 26 µm, at the lower edge of the design box, 25 µm). A 30 GHz gap this low in frequency needs strong inversion breaking, so the hole sizes are nearly fixed. The freedom left over is in the shape harmonics (c_3, s_3, c_6, s_6), which is where the designs differ: that is the non-uniqueness, made visible. Allowing smaller holes (a wider design box) would give the optimiser more room.

11. How to run everything, and the COMSOL side

11.1 Commands

All scripts run from the Project folder with the project's virtual environment (`.venv\Scripts\activate`). Times are for this laptop (12 cores, RTX 3050).

```
python src/utils/process_bands.py                          # raw COMSOL CSVs -> bands.csv, band_edges.csv (once)
python src/shapes/validate_pwe.py                          # fit eps, check PWE vs 128 COMSOL cells   (~1 min)
python src/shapes/gen_dataset.py --kind fourier --n 4000  # freeform C3 cells, PWE labels           (~3 min)
python src/shapes/gen_dataset.py --kind library --n 2000  # mixed-shape cells                         (~1 min)
python src/shapes/add_comsol_dataset.py "<csv>" --family circle   # add a COMSOL sweep (section 11.4)
python src/shapes/check_mk_shortcut.py                     # band edges only at M and K? (6.4)      (~3 min)
python src/shapes/run_optionb.py --showcase 300 330       # surrogate, multi-fidelity, inverse, gallery (~15 min)
python src/shapes/make_figures.py                          # figures for this guide
python src/shapes/build_guide.py                           # this PDF
```

11.2 Building a Fourier-hole model in COMSOL

To verify designs (or to generate high-fidelity freeform data), the hole outline must be driven by global parameters. Work on a copy of the existing model:

1. **Definitions → Parameters:** add rA0, c3A, s3A, c6A, s6A, rB0, c3B, s3B, c6B, s6B (all in [um]); the hole centres xA, yA, xB, yB as used by your model; and thA = the direction from hole A to one of its B neighbours (30[deg] in this guide's convention: A = (a/2, a/(2√3)), B = (a, a/√3)), thB = thA + 180[deg].
2. **Geometry → More Primitives → Parametric Curve** (hole A): parameter s from 0 to 2*pi, with

   ```
   x: xA + (rA0 + c3A*cos(3*s) + s3A*sin(3*s) + c6A*cos(6*s) + s6A*sin(6*s))*cos(s + thA)
   y: yA + (rA0 + c3A*cos(3*s) + s3A*sin(3*s) + c6A*cos(6*s) + s6A*sin(6*s))*sin(s + thA)
   ```
3. **Conversions → Convert to Solid** on the curve. Repeat steps 2–3 for hole B with the B parameters and thB.
4. If your cell cuts holes at its boundary (the slides' hexagonal cell has the holes on its corners), copy each hole to every equivalent corner with **Transforms → Copy** (displacements = lattice vectors), then subtract all holes from the cell with **Booleans → Difference**. Keep the Floquet periodicity, materials and mesh settings unchanged.
5. **Sanity check:** set all c and s to 0 (circles) and compare with the round-cell data; or set c3 = −0.160·r0 and r0 = 0.339·L to approximate a rounded triangle and compare with the old data.
6. **Eigenfrequency step:** ask for 4 eigenfrequencies searched around a frequency below band 1 (e.g. 200 GHz), so bands 1 and 2 are always among them; the parsers keep the two lowest per k. Use only k = 1 and k = 2 (chapter 6.4).

11.3 Verifying the gallery designs

run_optionb.py writes `data/pwe/showcase_for_comsol.txt` in COMSOL's Load-from-File format (10 shape parameters in metres plus k, every design listed once per k) and `data/pwe/showcase_designs.csv` with the PWE band edges to compare against. In **Study → Parametric Sweep** choose *Specified combinations*, *Load from File*, set the unit of the ten length parameters to m, run, then **Results → Derived Values → Global Evaluation** of emw.freq (GHz) and export the table. Expect agreement within ~1–2 GHz; larger differences mean the PWE calibration does not carry over to freeform shapes and the multi-fidelity correction needs freeform COMSOL points.

11.4 Adding a new COMSOL sweep (how the round-cell data went in)

```
python src/shapes/add_comsol_dataset.py "data/comsol/r=a_divby(3sqrt(3))_(Parametric sweep at k=1,2).csv"
--family circle

python src/shapes/check_comsol_rerun.py "data/comsol/parameter_sweep_values (Varied L1, L2 and r = 2Lby
3sqrt(3)) (k=1,2).csv" --corner 1
```

The first command reads the per-mesh-node export in chunks (columns X, Y, r1, r2, k, λ, freq; use --usecols if different), keeps the two lowest eigenfrequencies per k, writes `data/comsol/extra_circle.csv` in the shared format and prints the PWE–COMSOL error. The second compares a k = 1, 2 re-run with the old full-path sweep of the same cells. Then rerun run_optionb.py, make_figures.py and build_guide.py: new cells automatically join the COMSOL test set and the multi-fidelity correction. The same route works for any future sweep (for example the gallery designs) with --family rounded, sharp or circle; freeform designs need a parser for the 10-parameter table.

11.5 Code map (src/shapes)

File	Purpose
geometry.py	lattice, hole families, C3 Fourier descriptor, validity checks
pwe.py	plane-wave-expansion band solver
validate_pwe.py	fit ε, compare with both COMSOL datasets
gen_dataset.py	sample valid cells, label with PWE, check the M/K shortcut
dataset.py	one table for COMSOL + PWE data with the 18 features
add_comsol_dataset.py	add a new COMSOL sweep (e.g. circles)
check_comsol_rerun.py	compare a k = 1, 2 COMSOL re-run with an old full-path sweep
check_mk_shortcut.py	check that band edges sit only at M and K (section 6.4)
models.py	surrogate ensemble, validity penalty, tandem, CVAE, multi-start
run_optionb.py	all experiments, gallery, COMSOL export
make_figures.py, build_guide.py	figures and this PDF

12. Limitations — say these before an examiner does

- **Low-fidelity labels.** The 6,000 cells are labelled by the 2D PWE solver, calibrated with one number (ε) on triangles only. Freeform shapes are not yet confirmed in COMSOL; the gallery file is the first check.
- **2D is itself an approximation.** Both COMSOL and PWE treat the 200 µm slab as a 2D crystal with an effective permittivity. A real slab has a light cone, radiation loss and an effective index that depends on frequency and on the hole pattern. Final designs need a 3D (slab) check, as in the group's device simulations.
- **Gap is not topology.** A C3-symmetric cell with unequal holes and a gap at K is expected to be valley-Hall, but the valley Chern sign, band ordering and the existence of edge states for a chosen interface are not computed yet.
- **Fabrication rules** (25 µm walls, 8 µm corners) are placeholders; replace them with your process limits.
- **Targets were drawn from the same distribution as the data**, so they are reachable. A real request can be unreachable; the surrogate's uncertainty and the nearest achievable target should be reported then.
- **Correction learned from simple shapes only.** The multi-fidelity correction has seen rounded triangles, sharp triangles and circles; applying it to freeform shapes is extrapolation until freeform COMSOL data (the gallery designs) are added.
- **Several splits, one dataset.** The benchmark was repeated with different random splits (chapter 10, repeatability table), but all runs use the same 6,000 PWE cells and the same PWE 'truth'; COMSOL verification of freeform designs is still missing.

13. What to do next

Before the November evaluation

1. Confirm the permittivity in the COMSOL model (chapter 6) and the hole construction (chapter 3.3).
2. Run the gallery designs in COMSOL (chapter 11.3) — even 4–8 designs turn 'PWE says' into 'COMSOL confirms'.
3. Run 2–3 circle cells on the full k-path once in COMSOL (section 6.4) — the round-cell data is already in (chapter 3.5).
4. Presentation storyline: problem → stage-1 audit (one-to-one, GP wins) → why freeform shapes (non-uniqueness) → symmetry rule and Fourier descriptor → validated fast solver → dataset → surrogate and learning curve → four inverse methods compared with physics verification → gallery of different shapes for one target → limitations and plan.

Next semester

- **COMSOL in the loop (active learning):** pick the next COMSOL runs where the surrogate is most uncertain or where the inverse designs land (MPh scripts in src/comsol already run COMSOL from Python).
- **Proper multi-fidelity model** (co-kriging / Δ-learning network) trained on PWE + COMSOL across all shape families.
- **Topology-aware design:** compute the valley Chern number from the PWE eigenvectors (Berry curvature near K) and add it as a constraint; then design the VPC-A/VPC-B interface and its edge-state band.
- **Richer shapes** (harmonics 9 and 12, or free-form pixels with a diffusion prior, as in Zhang et al. 2026) and a 3D slab check; fabrication and THz measurement of one design.

14. Glossary

Term	Meaning in this project
unit cell	smallest repeating tile of the crystal; here a rhombus with two holes (A, B)
Bloch / Floquet	wave = plane wave × periodic function; COMSOL's periodic boundary with phase e^(ik·R)
k-point, k-path	wave vector; the band diagram follows Γ → M → K → Γ (k = 0, 1, 2, 3)
band edge	f_lower = highest point of band 1; f_upper = lowest point of band 2
band gap Δf	f_upper − f_lower; negative means the bands overlap (no complete gap)
Dirac point	point where bands 1 and 2 touch at K when both holes are equal (C6 symmetry)
valley, valley Chern number	K and K′; topological index ±1/2 whose sign flips when the holes are swapped
C3 / C6 symmetry	unchanged by 120° / 60° rotation
C3 Fourier descriptor	r(θ) = r0 + Σ c_n cos nθ + s_n sin nθ with n = 3, 6, 9, 12; 9 numbers per hole
PWE	plane-wave expansion: Maxwell's equations as a matrix eigenproblem in a plane-wave basis
effective index / ε	a 2D stand-in for the slab: ε ≈ 8.9 (n ≈ 2.99) instead of bulk 11.7
fidelity	how close a simulator is to reality; PWE low, COMSOL (2D) higher
surrogate / forward model	network or GP that predicts band edges from shape in milliseconds
inverse model	anything that goes from target band edges to a shape
non-uniqueness	many shapes give the same target; makes direct inverse regression fail
tandem network	inverse network trained through a frozen forward network (loss on the response)
CVAE	conditional variational autoencoder; samples many designs for one target
ensemble	several networks trained from different seeds; their spread estimates uncertainty
Gaussian process (GP)	Bayesian regression with a kernel; strong on small data, gives uncertainty
multi-fidelity correction	learn COMSOL − PWE and add it to cheap predictions
validity penalty	smooth cost that is zero only for fabricable cells
MAE	mean absolute error, here always in GHz

15. Test yourself

Q. Why did the tandem network not beat a Gaussian process in stage 1?

A. With two design numbers and two targets the map was one-to-one and smooth; there was no non-uniqueness to resolve, and 61 points favour a GP.

Q. Why can a hole not have a cos 2θ term?

A. It is not invariant under 120° rotation, so it breaks C3 and the valley physics.

Q. What are f_lower and f_upper and where are they found?

A. Max of band 1 (always at K here) and min of band 2 (at M for rounded triangles, circles and most freeform cells; at K for most sharp triangles); the gap is their difference.

Q. Why may COMSOL be run at only k = 1 and 2, and when could that fail?

A. The gap opens at K and band 2 bottoms out at M or K; this was checked on all COMSOL cells and on 700+ PWE cells over the whole irreducible zone. It is not a theorem: a new hole family could put an edge between K and M (as circles do in some 3D slabs), so check a few cells of each new family on the full path.

Q. Why does swapping holes A and B not change the band edges?

A. Swapping is equivalent to inversion of the crystal; together with time reversal the frequencies are unchanged — but the valley Chern number flips.

Q. Why fit ε when the slides say 11.7?

A. All frequencies were 13% off by the same factor; one permittivity explains it (likely an effective index in the 2D COMSOL model). After fitting, both triangle datasets agree within ~1 GHz — and so does the circle dataset, which was not used in the fit.

Q. What does the validity penalty do in the tandem and multi-start methods?

A. It pushes designs back into the fabricable region during training/optimization; an exact check is still applied afterwards.

Q. Why verify with the PWE solver rather than with the surrogate?

A. An optimizer can exploit surrogate errors; only an independent simulation shows whether the target is really met.

Q. What would make the multi-fidelity correction trustworthy for freeform shapes?

A. COMSOL runs of freeform (and circle) cells, e.g. the gallery designs, added to the correction's training data.

Q. How do you move a design to a different 6G channel?

A. Scale the lattice constant and the holes by s; all frequencies divide by s (2D, non-dispersive).

16. References

1. D. Liu, Y. Tan, E. Khoram, Z. Yu, "Training deep neural networks for the inverse design of nanophotonic structures", ACS Photonics 5, 1365 (2018) — the tandem network. arxiv.org/abs/1710.04724
2. J. D. Joannopoulos, S. G. Johnson, J. N. Winn, R. D. Meade, *Photonic Crystals: Molding the Flow of Light*, 2nd ed., Princeton (2008) — band structure, master equation, plane waves. ab-initio.mit.edu/book
3. K. M. Ho, C. T. Chan, C. M. Soukoulis, "Existence of a photonic gap in periodic dielectric structures", Phys. Rev. Lett. 65, 3152 (1990) — inverse-matrix rule for PWE.
4. Y. Yang et al., "Terahertz topological photonics for on-chip communication", Nature Photonics 14, 446 (2020) — the a = 242.5 µm silicon VPC platform.
5. T. Ma, G. Shvets, "All-Si valley-Hall photonic topological insulator", New J. Phys. 18, 025012 (2016).
6. M. I. Shalaev, W. Walasik, A. Tsukernik, Y. Xu, N. M. Litchinitser, "Robust topologically protected transport in photonic crystals at telecommunication wavelengths", Nature Nanotechnology 14, 31–34 (2019), doi:10.1038/s41565-018-0297-6; arXiv:1712.07284 — triangular vs circular holes, band 2 at M, 2D COMSOL model with n_{eff} = 2.965 (sections 6.3, 6.4).
7. J. M. Harrison, P. Kuchment, A. Sobolev, B. Winn, "On occurrence of spectral edges for periodic operators inside the Brillouin zone", J. Phys. A: Math. Theor. 40, 7597–7618 (2007); arXiv:math-ph/0702035 — band edges need not lie on the zone boundary.
8. R. V. Craster, T. Antonakakis, M. Makwana, S. Guenneau, "Dangers of using the edges of the Brillouin zone", Phys. Rev. B 86 (2012).
9. F. Maurin, C. Claeys, E. Deckers, W. Desmet, "Probability that a band-gap extremum is located on the irreducible Brillouin-zone contour for the 17 different plane crystallographic lattices", Int. J. Solids Struct. 135, 26–36 (2018), doi:10.1016/j.ijsolstr.2017.11.006.
10. A. Kumar et al., "Terahertz topological photonic integrated circuits for 6G and beyond: a perspective", J. Appl. Phys. 132(14) (2022).
11. K. Sohn, H. Lee, X. Yan, "Learning structured output representation using deep conditional generative models", NeurIPS (2015) — conditional VAE.
12. W. Ma, F. Cheng, Y. Xu, Q. Wen, Y. Liu, "Probabilistic representation and inverse design of metamaterials based on a deep generative model with semi-supervised learning", Adv. Mater. 31, 1901111 (2019).
13. B. Lakshminarayanan, A. Pritzel, C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles", NeurIPS (2017).
14. C. E. Rasmussen, C. K. I. Williams, *Gaussian Processes for Machine Learning*, MIT Press (2006). gaussianprocess.org/gpml
15. M. C. Kennedy, A. O'Hagan, "Predicting the output from a complex computer code when fast approximations are available", Biometrika 87, 1 (2000) — multi-fidelity modelling.
16. R. E. Christiansen, O. Sigmund, "Inverse design in photonics by topology optimization: tutorial", JOSA B 38, 496 (2021) — in your Reading Material folder.
17. Label-efficient generative inverse design and design-rule discovery in freeform topological photonics, arXiv:2609.28401 (2026) — C3-symmetric freeform valley crystals with a diffusion prior.
18. Valley photonic crystal design trading band-gap width against valley Chern number, arXiv:2601.22958 (2026).
19. Inverse design of dual-band valley-Hall photonic crystals, arXiv:2503.22206 (2025).
20. IEEE Std 802.15.3d-2017 — 252.72–321.84 GHz wireless links.
21. COMSOL 6.4 documentation: Parametric Curve, Parametric Sweep (Load from File), Global Evaluation. doc.comsol.com